

Machine Learning

Lecture 1: Overview



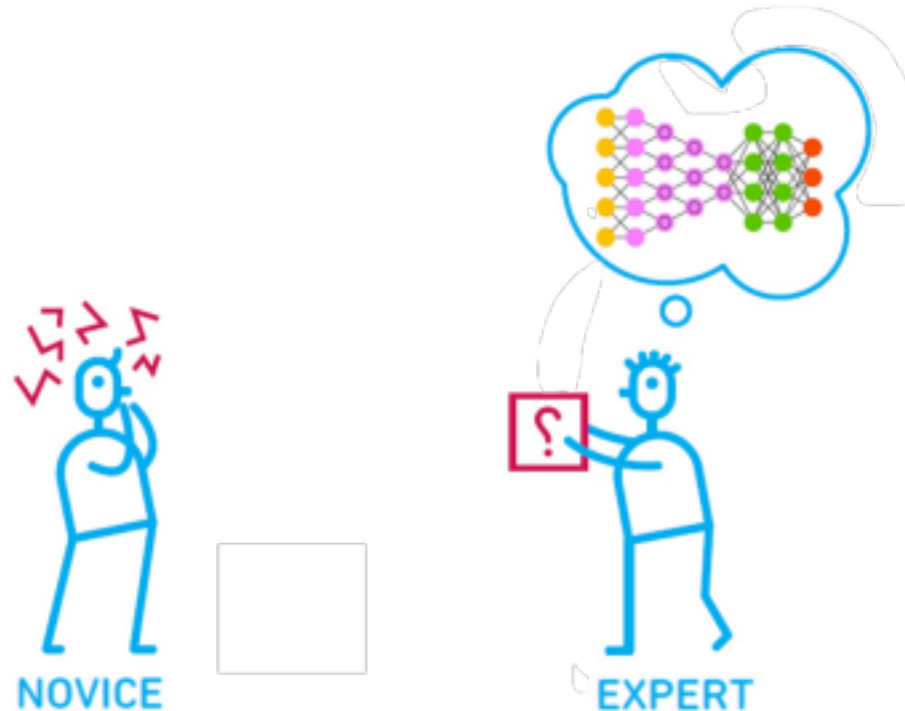
hunkim+ml@gmail.com

Goals

- Basic understanding of machine learning/deep learning
- PyTorch implementation skills

- Zero to All!

- Basic algebra + probability
- Basic python



What is ML?

Adapted the Illustration by Oliver Caviglioli



What is Human

Intelligence?



What is Human
Intelligence? *What to eat
for lunch?*



What is Human
Intelligence? *What to eat
for lunch?*



Information

What is Human
Intelligence? *What to eat
for lunch?*

Infer



Information

What is
Human Intelligence? *What
to dress?*



Information_{Infer}

What is
Human Intelligence? *What
is this picture?*



Image information



Prediction

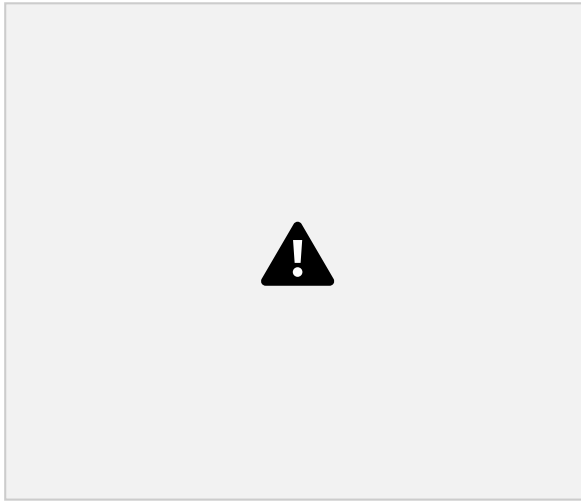
CAT

What is Human
Intelligence? *What is this
number?*



What is Human
Intelligence? *What would be the grade
if I study 4 hours?*

4 hours



information Prediction

?

points

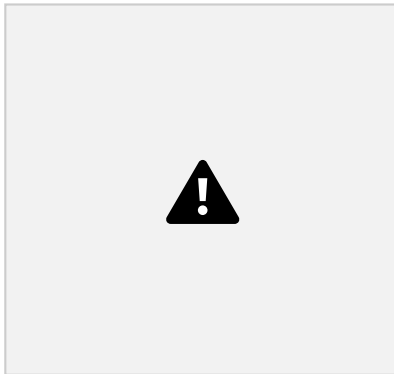
Machine Learning

What to dress?



Machine Learning

What is this picture?

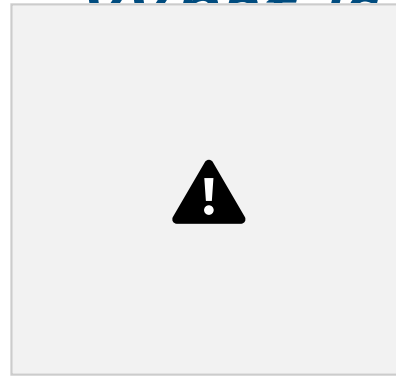
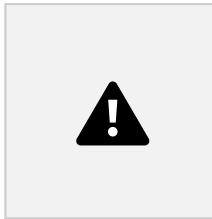


CAT

Image information Prediction

Machine Learning

What is this number?



2 Image information Prediction

Machine Learning

What would be the grade if I study 4 hours?

4 hours  ?
points

information Prediction

Machine Learning

Machine needs lots of training 

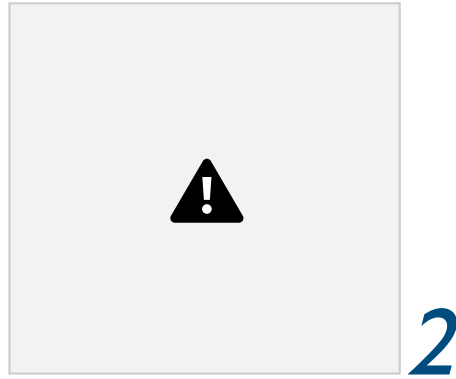
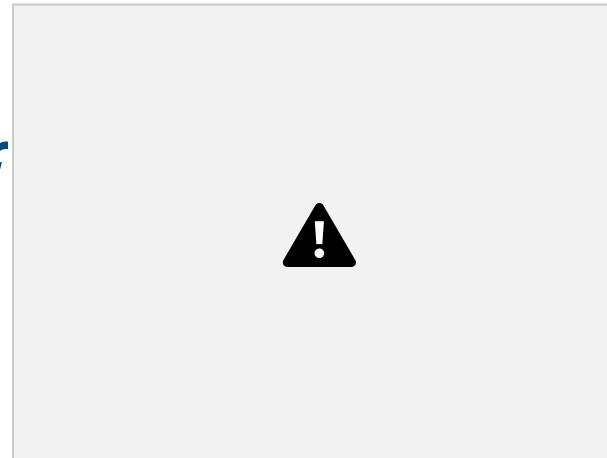
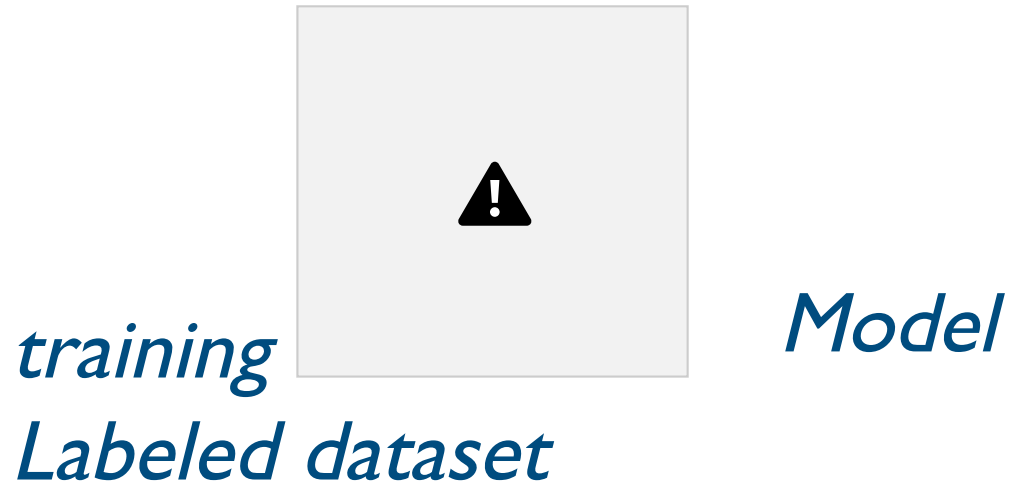


Image information Prediction

Machine Learning

Machine needs lots of training





Machine Learning
Predict (test) with trained



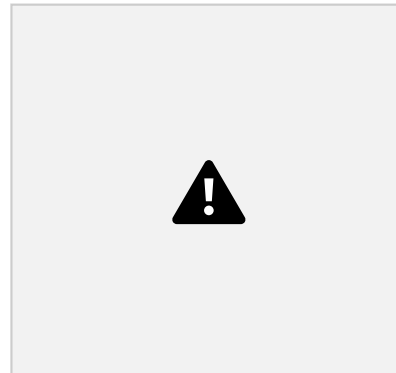


*Trained
model*

Machine Learning

Predict (test) with trained

model





*Test
dataset*

model

Trained

Machine Learning

Predict (test) with trained

model

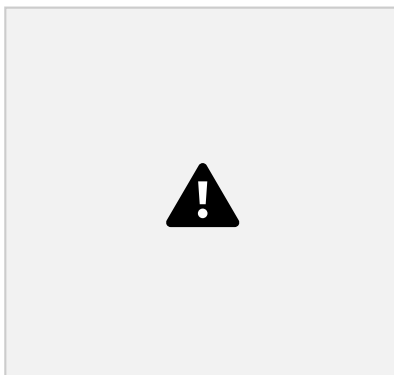


Image information Prediction

*Test
dataset*

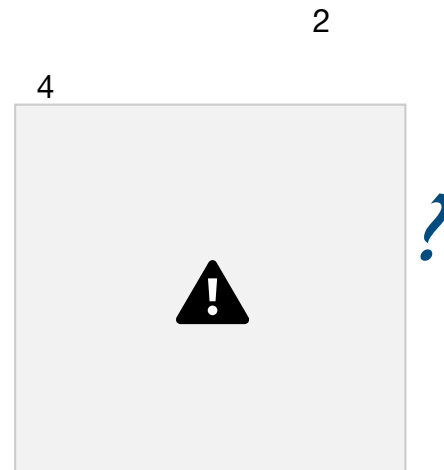
model

Trained

Machine Learning

What would be the grade if I study 4 hours?

4 hours



points
Prediction

dataset

Training

? Test dataset

Deep Learning?



Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville

Why We Care?



More demos at

<https://ml-showcase.com/>

Why We Care as Developer?



Deep Learning by [Ian Goodfellow](#), [Yoshua Bengio](#), [Aaron Courville](#)

Why We Care as
Developer?

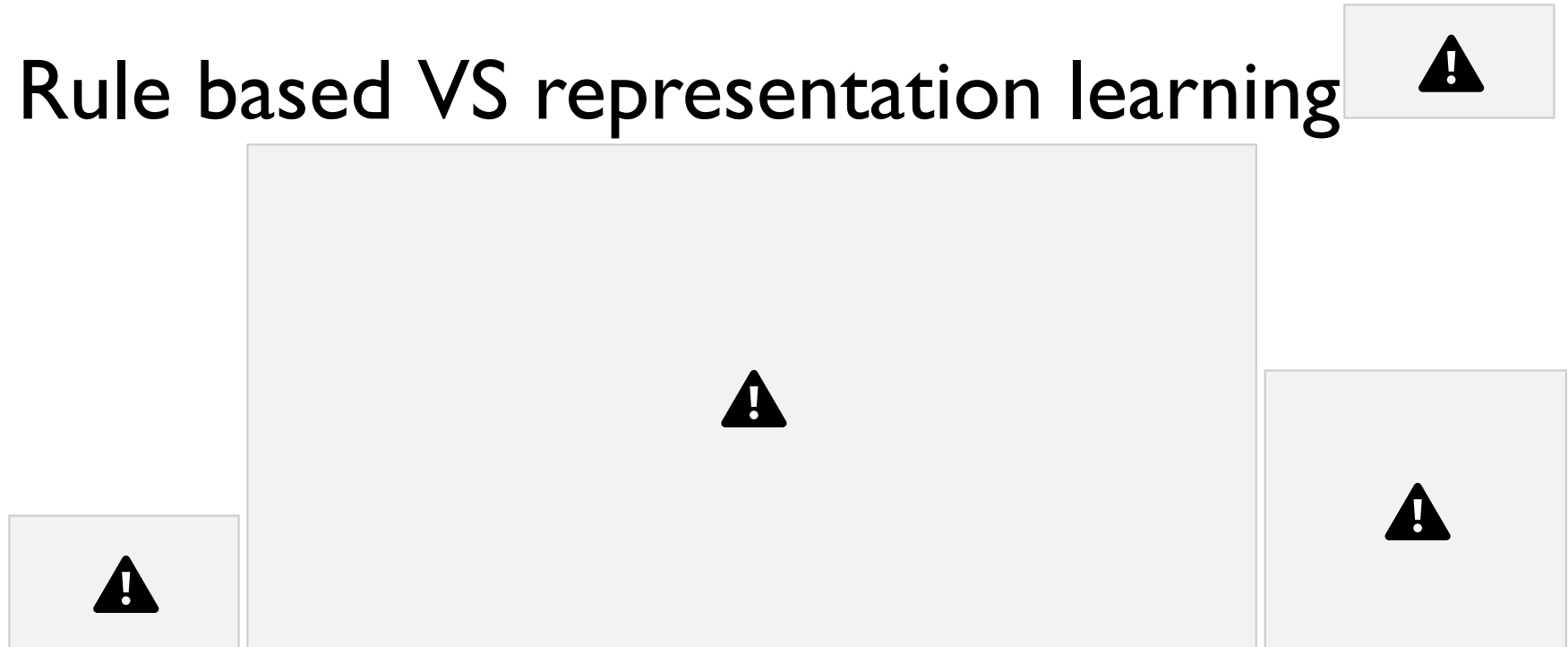


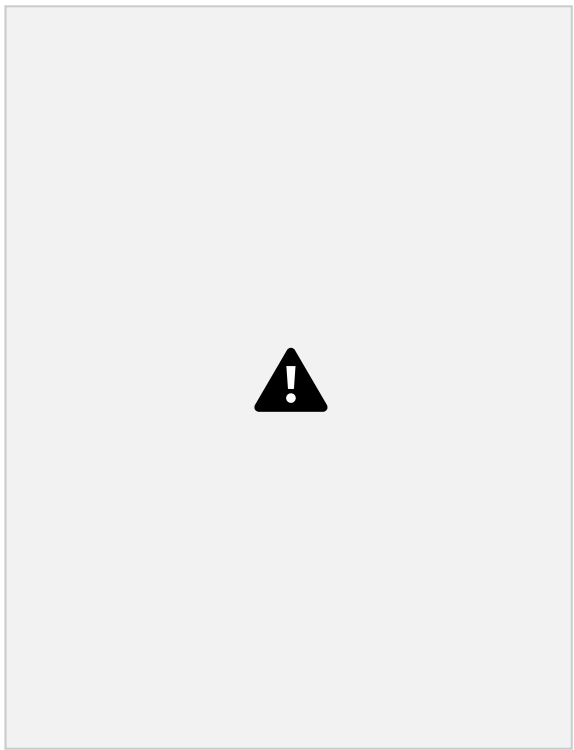
Deep Learning by [Ian Goodfellow](#), [Yoshua Bengio](#), [Aaron Courville](#)

Why We Care as
Developer?



Rule based VS representation learning





PyTorch is a python package that provides two high-level features:

- Tensor computation (like numpy) with strong GPU acceleration
- Deep Neural Networks built on a tape-based autograd system

Why



<http://pytorch.org/about/>

- More Pythonic (imperative)
 - Flexible
 - Intuitive and cleaner code
 - Easy to debug
- More Neural Networkic
 - Write code as the network works
 - forward/backward

Install





<http://pytorch.org>
g

Exercise I-I:
Install PyTorch on your computer!

```
09:40 $ python3
```

```
Python 3.6.2 (v3.6.2:5fd33b5926, Jul 16 2017, 20:11:06 [GCC 4.2.1  
(Apple Inc. build 5666) (dot 3)] on darwin Type "help", "copyright",  
"credits" or "license" for more information. >>> import torch
```

```
>>> print(torch.__version__)
```

```
0.2.0_3
```

```
>>> # Happy!!
```

Topics

- Linear, Logistic, softmax models
- DNN: Deep Neural Net
- CNN: Convolutional Neural Net
- RNN: Recurrent Neural Net

- Write everything in PyTorch



Lecture 2: Linear Model